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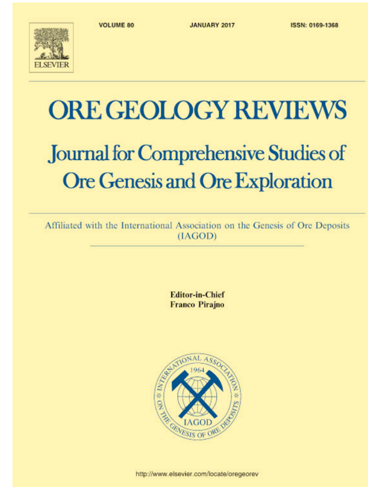

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Coloration mechanism of Fe in sphalerite based on LA-ICP-MS and DFT: A case study of the Huize super-large Ge-rich Ag–Pb–Zn deposit in Sichuan–Yunnan–Guizhou

Yanhong Liu[a], Runsheng Han[a*], Yan Zhang[a*], Yi Chen[a], Jianbiao Wu[a]

[a] *Kunming University of Science and Technology, Southwest Institute of Geological Survey, Geological Survey Center for Nonferrous Metals Resources, Kunming 650093, China*

Abstract: The Fe content is a crucial factor influencing the color of sphalerite. This study focuses on sphalerite from the Huize super-large Ge-rich Ag–Pb–Zn deposit in the Sichuan–Yunnan–Guizhou border region. Based on LA-ICP-MS in-situ analysis and quantum chemical calculations using Density Functional Theory (DFT), we investigated the Fe–Zn substitution patterns and their impact on sphalerite color, identified the isomorphous and interstitial doping modes of iron in sphalerite, and further elucidated the coloration mechanism and mineralization process of sphalerite. The research shows that black sphalerite from the Baizuo Formation in this deposit has the highest Fe content, while red-brown sphalerite from the Dengying Formation has the highest Fe content among its counterparts. When the number of Fe atoms in the ZnS electronic structure exceeds six, the reflection coefficient in the red-brown spectral range is significantly higher than that in other visible light bands, while the absorption coefficient is lower. The isomorphous substitution and interstitial insertion of Fe in sphalerite are key factors leading to its color variation. Fe predominantly enters the (110) and (100) cleavage planes in the form of isomorphous substitution, forming isomorphous mixed crystals, with the (110) cleavage plane being more sensitive to Fe substitution. The study suggests that the mineralization temperature corresponding to different mineralization stages affects the Fe content in sphalerite, which in turn determines its color presentation. This further supports the evolution process of the ore-forming fluid from medium–to–high temperature, and low salinity to medium temperature, and medium salinity, and finally to low temperature, and medium–to–low salinity. This paper provides a new theoretical perspective for revealing the coloration mechanism of sphalerite.

Keywords: Coloration mechanism; LA-ICP-MS; DFT; Sphalerite; Huize Ge-rich Ag–Pb–Zn deposit

1. Introduction

The varying contents of certain elements in metallic minerals hold different scientific significance. Iron, which accounts for 5% of the Earth's crust, is one of the primary elements that determine the color of minerals (Kurt, 1978). The color of sphalerite can undergo significant changes depending on the type and content of trace elements. Some of the zinc in its crystal lattice can be replaced by iron. As the content of Fe^{2+} increases, the color of sphalerite changes from light to dark (Li et al., 2020; Yang et al., 2022), and its transparency correspondingly decreases from transparent to translucent to opaque. The luster also changes from adamantine to semimetallic. If the iron content is high, causing it to appear dark in color, it is referred to as iron sphalerite or black sphalerite (Liu et al., 1984; Xiong, 2013; Wang, 2020). The important factors that determine the color of minerals mainly include electron transitions of metal ions, electron transfers between ions, color center coloring, electron transitions between energy bands, and physical optics. Generally speaking, the color of each mineral is caused by one or more of these factors (Jia et al., 2020). In the study of the optical properties of minerals, both the reflection coefficient and the absorption coefficient are important parameters for describing the interaction between minerals and light. Based on the optical response of the medium at all phonon energies $E=h\omega$, it can be described using the complex dielectric function $\varepsilon(\omega)=\varepsilon_1(\omega)+i\varepsilon_2(\omega)$. By combining the Kramers-Kronig relations, the real part of the dielectric function ε_1 can be determined, and then the reflection coefficient $R(\omega)$ and the absorption coefficient $\alpha(\omega)$ can be derived. The two coefficients can independently describe different ways in which light interacts with the material. The reflection coefficient describes the reflection behavior of light at the interface, while the absorption coefficient describes the attenuation behavior of light as it propagates within the material. By calculating and simulating the impact of iron atoms at different doping positions on the electronic structure and optical properties of sphalerite, we can construct an intrinsic relationship model between iron content and the color of sphalerite. Furthermore, we can analyze the correlation between the reflection coefficient and absorption coefficient of the sphalerite cleavage plane, as well as their corresponding relationship with color(Lucarini et al.,2005; Wu et al.,2024).

The Sichuan–Yunnan–Guizhou Pb–Zn polymetallic mineralization zone is one of the 22 key mineral exploration zones in China's resource planning, as well as the largest Pb–Zn industrial base and a global production base for germanium. It is widely distributed with globally rare carbonate-hosted, non-magmatic, hydrothermal (CNHT) Ge-rich Pb–Zn deposits that are clearly controlled by fault-fold tectonic systems, known as the Huize-type (HZT) Pb–

Zn deposits (Han et al., 2022, 2023, 2024; Wu et al.,2024a). Among these, the ore mineral sphalerite is the most dominant, exhibiting diverse colors, compositions, and structural differences. The Huize super-large Ge-rich Ag–Pb–Zn deposit is one of the typical representatives of this region, characterized by exceptionally high Pb–Zn grades, large reserves, and rich contents of the rare element germanium and the precious metal silver. The sphalerite in this deposit mainly consists of three generations, corresponding to three mineralization stages. From the early to the late generation, its color changes from dark brown to brown, then to brownish, and finally to light yellow (Liu et al., 2022; Niu et al., 2024). Previous studies have shown that the color of sphalerite is closely related to the mineralization stage. In the early stage, sphalerite is dark brown with a medium–to coarse-grained euhedral granular structure. In the main stage, sphalerite is brown with a coarse-grained granular structure and often occurs as veins with galena and pyrite. In the late stage, sphalerite is light yellow with a medium-grained structure and obviously replaces gangue minerals (Zhang et al., 2015; Liu et al., 2024; Liu et al., 2024; Cook et al., 2009; Ye et al., 2011; Wei et al., 2019; Li et al., 2020; Zhang et al., 2022; Xing et al., 2022). However, there are few reports on the coloration mechanism of sphalerite.

Therefore, this paper takes the Huize super-large Ge-rich Ag–Pb–Zn deposit as the research object, employing LA-ICP-MS in-situ analysis and density functional theory calculations. We extract the trace element content and distribution patterns of sphalerite, analyze the correlation between sphalerite color and iron content, clarify the Fe–Zn substitution pattern and its impact on sphalerite color, and identify the isomorphous and interstitial doping modes of iron in sphalerite. Furthermore, we aim to reveal the coloration mechanism of sphalerite and the mineralization process. From a new perspective of studying the formation mechanism of sphalerite color, this research provides important information for unveiling the mineralization mechanism and evolution process of this type of deposit.

2. Geological setting

The Sichuan–Yunnan–Guizhou Pb–Zn polymetallic metallogenic region is located at the southwestern margin of the Yangtze Block, at the junction of the Tethyan metallogenic domain and the Circum-Pacific metallogenic domain (Fig.

1) (Han et al., 2012, 2019). This region is bounded by deep faults: to the west by the NS-trending Anninghe–Lvzhijiang Fault, to the north by the Kangding–Yiliang–Shuicheng Fault, and to the south by the Mile–Shizong–Shuicheng Fault (Wang et al., 2004) (Fig. 1a). The Huize Ge-rich Ag–Pb–Zn deposit is one of the most representative super-large Ge-rich Ag–Pb–Zn deposits in the region. The main ore-controlling structures are faults and folds, with the three NE-trending compressional–torsional faults of Kuangshanchang, Qilinchang, and Yinchangpo serving as the main structures, respectively controlling the Kuangshanchang, Qilinchang, and Yinchangpo deposits. The exposed strata in the mining area mainly include the Sinian (Ediacaran), Cambrian, Devonia, Carboniferous, and Permian(Fig. 1b, c). Among them, the ore-bearing strata are the gray to grayish-white thick-bedded medium–to–coarse-grained dolomite of the Baizuo Formation (C_1b) of the Lower Carboniferous and the light gray to gray medium–to–thick-bedded siliceous dolomite of the Dengying Formation (Zb*dn*) of the Upper Sinian (Zhang et al., 2023). The ore bodies are hosted within interlayer fault zones, trending NE, dipping SE, with a dip angle of 50° to 55°; the ore bodies extend up to 400m in length and exceed 1700m in vertical depth, occurring in a bed-like, sac-like, or lenticular shape etc.. The ore minerals include sphalerite, galena, pyrite, and a small amount of chalcopyrite, while the gangue minerals are dolomite, calcite, followed by quartz, barite, gypsum, and illite. The ore texture is predominantly dense and massive, with a few veins, irregular bands, and stockwork textures observable. The ore structure is mainly xenomorphic to euhedral granular and metasomatic, with features such as common boundaries, interstitial filling, crumpling, crushing, and inclusion. The ore grade is rich, with an average Pb+Zn grade exceeding 25%.

Based on the typical characteristics of the deposit, ore texture and structure, mineral paragenetic association, and vein cross-cutting relationships, combined with previous research, the hydrothermal mineralization period is divided into four stages (Fig. 2): (I) Sphalerite–pyrite stage, (II) Sphalerite–galena–pyrite stage, and (III) Pyrite–galena–sphalerite–calcite stage, (IV) Pyrite–calcite stage.

Combined with previous studies, Wang et al. (2008) classified the sphalerite of Huize Pb–Zn deposit into dark and light colors, according to the negative correlation before Fe and Cd, Ge, Ga, indicating that the three

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dispersed elements substitute Fe in and out of sphalerite by isomorphous substitution, and the sphalerite becomes lighter in color with the reduction of the Fe content(Wang et al. 2008); Wang (2017) classified the color of sphalerite into colorless, light yellow, and light orange through the identification of the specimen under the microscope, orange-red, sphalerite is seen in oxidized ore as dark brown, and in the shallow part of the ore body is dominated by light-colored sphalerite, and the deep sphalerite color is gradually deepened(Wang, 2017); Zhang et al. (2015) believes that sphalerite is the main ore mineral in Huize Pb–Zn deposits, and it is available from light yellow, yellow-brown, brown, brown, and black-brown, and it is mostly co-occurring with galena and pyrrhotite(Zhang et al.); Li et al. (2020) identified that primary sphalerite is black, brown, light yellow, etc. Light-colored sphalerite is mostly found in the contact area between sphalerite and calcite(Li et al. 2020); Li (2022) divided sphalerite into three generations, the first generation of sphalerite is red brown-dark brown and symbiotic with pyrite, the second generation of sphalerite is red brown-light red brown and is closely symbiotic with galena, and the third generation of sphalerite is yellowish brown and is dipped and stellate distributed in calcite, and the grain size of sphalerite is changed from the early stage of metallogenesis to the late stage of metallogenesis(Li, 2022). From the early stage of mineralization to the late stage, sphalerite becomes finer in grain size and lighter in color (Wang et al. 2008; Zhang et al.,2015; Wang, 2017; Li et al. 2020; Li, 2022).

3. Material and methods

The study focuses on iron as the research object to determine the relationship between iron content and the color intensity of sphalerite. The six samples used in this test and research were collected from four different colored sphalerites at the 1031m and 1404m levels of the ore-bearing strata in the Huize Pb–Zn deposit. The sampling locations and characteristics are shown in Table 1.

3.1 In situ LA-ICP-MS trace elements analysis

The in-situ micro-area trace element content testing was completed using LA-ICP-MS at Guangzhou Tuoyan Testing Technology Co., Ltd. During the analysis and processing of in-situ micro-area trace element content, glass standard materials NIST SRM610 and MASS-1 were used for multi-external standard total normalization correction (Liu et al., 2008; Jochum et al., 2011), and the proportional standard material SRM 612 was employed as a monitoring sample (Wilson et al., 2002; Danyushevsky et al., 2011). Offline processing of the analytical data, including the selection of sample and blank signals, correction for instrumental sensitivity drift, and calculation of element content, was accomplished using the 3D Trace Elements DRS mode of the IOLITE software (Paton et al., 2011). For detailed testing methods, please refer to Zhang et al. (2022).

3.2 LA-ICP-MS trace-element mapping of magnetite

The test was conducted at the Guangzhou Tuoyan Testing Technology Center, utilizing the combination of an NWR 193 nm ArF Excimer laser ablation system and an iCAP RQ (ICPMS) to determine the trace element profiles of the samples. Two sets of test parameters were employed: the first set involved a laser fluence of 5 J/cm^2, a repetition rate of 15 Hz, a spot size of 5 μm, and a laser scanning speed of 20 μm/s; the second set featured a laser energy density of 5 J/cm^2, a repetition rate of 15 Hz, a spot size of 8 μm, and a laser scanning speed of 25 μm/s. Following the analysis of two sulfide standard blocks (one NIST 610 and one NIST 612), 30 to 50 unknown samples were analyzed. All isotopes were corrected using the external standard method.

3.3 Theoretical Models and Computational Methods

The experimental parameter calculations were based on the quantum chemistry calculation program CASTEP, which utilizes Density Functional Theory (DFT). The interactions between ions and valence electrons were described using Projector Augmented Wave-Pseudopotential (PAW-PP), with the electron wave functions represented by plane wave functions. The plane wave cutoff energy was set to 450 eV, and the Brillouin zone integration grid followed a 4×4×4 Monkhorst-Pack scheme (Kresse, 1996; Martin, 2004; Tran et al., 2009). The convergence criterion for the iterative process was 5×10^{-4} eV/atom, and the convergence standard for the interatomic interaction force was set to 0.1 eV/0.1 nm. To investigate the impact of Fe on the electronic structure of sphalerite, a 2×2×2 supercell was constructed, with Fe substituting for Zn. Using the Interstitial-substitutional (I-S) diffusion mechanism, substitutional doping and Interstitial doping were performed on the sphalerite supercell with Fe to obtain the corresponding doped systems. The reflection and absorption of relevant wavelength bands within the visible light range were then determined for the cleavage planes, allowing for the inference of the influence of different Fe contents on the color of sphalerite (Fig. 2, Table 2).

4. Results

4.1 Texture and sphalerite colors

The sphalerite produced in the two ore-bearing strata mostly appears as brown and red-brown, often with massive and vein-like structures (Fig. 4). Based on observations under transmitted light microscopy, four colors of sphalerite were identified: black, red-brown, yellow, and white. Additionally, zoning structures within the sphalerite were observed. This study found that the sphalerite zoning belongs to jump zoning, with alternating colors appearing in the zones and both light and dark colors present in the core (Fig. 4).

4.2 LA-ICP-MS spot analysis data

From the LA-ICP-MS analysis results (Table S1), it can be seen that the Fe content in sphalerite is relatively high, ranging from 87 ppm to 91632.5 ppm. The Fe content in black, red-brown, yellow, and white sphalerite ranges from 4079.6 ppm to 91632.49 ppm, 103.5 ppm to 73393.08 ppm, 2580 ppm to 28282.43 ppm, and 87 ppm to 1529.12 ppm, respectively. The highest average Fe content is found in black sphalerite, with a mean value of 35613.06 ppm; followed by red-brown sphalerite with an average Fe content of 29685.36 ppm, and yellow sphalerite with an average Fe content of 15225.89 ppm. The lowest average Fe content is observed in white sphalerite, with a mean value of 4710.47 ppm. Through data analysis and comparison of sphalerite from the two ore-bearing strata of the Baizuo Formation and the Dengying Formation, it is found that black sphalerite from the Baizuo Formation has the highest Fe content, with an average value of 44865.68 ppm, and the Fe content gradually decreases with lighter color from deep to shallow. In the Dengying Formation, red-brown sphalerite has the highest Fe content, with an average value of 30145.86 ppm. The peak Fe content is found in red-brown sphalerite, followed by black, and the lowest in white (Fig. 5).

4.3 LA-ICP-MS mapping

The sphalerite in this deposit mostly exhibits distinct zoning with varying colors, and the element distribution in sphalerite differs across different ore-bearing strata. To fully understand the distribution and variation characteristics of Fe in the sphalerite of this deposit, this study employed LA-ICP-MS mapping

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analysis to conduct geochemical testing on sphalerite samples. Compared to spot analysis, LA-ICP-MS trace element mapping analysis provides a more intuitive revelation of the correlation between Fe enrichment patterns and color in sphalerite.

The LA-ICP-MS trace element mapping distribution diagram reveals the distribution within sphalerite crystal grains. The mapping images show that the Fe content in sphalerite from the Dengying Formation of the Sinian System and the Baizuo Formation of the Carboniferous System has the following characteristics:

The concentration of Fe varies significantly across different color regions: the highest Fe content is found in black areas, followed by red-brown areas, while the Fe content in yellow areas is significantly lower than in red-brown areas, and the lowest Fe content is observed in white areas. This indicates a clear correlation between Fe content and color variation. The mapping images (Fig. 6) clearly show that, regardless of whether the sphalerite is from the Baizuo Formation or the Dengying Formation, the Fe content decreases as the color becomes lighter. The results of the mapping analysis are generally consistent with those of the LA-ICP-MS spot analysis for Fe. Compared to the distribution of Zn and S, Fe exhibits distinct zoning within the sphalerite.

4.4 DFT calculation results

Select the common cleavage planes (100) and (110) of sphalerite, and calculate their reflection coefficients and absorption coefficients. By calculating the reflection coefficients and absorption coefficients for each system in Table 2, we can understand the changes in the reflection coefficients of the (100) and (110) cleavage planes when Fe enters as isomorphous substitutions and interstitial atoms (Figs. 7 and 8).

(1)Calculation results of reflection coefficients for different doping conditions.

Calculating the substitutional doping case, when the solvation plane is the (100) crystal plane, the reflection coefficients of S-$Zn_{32}S_{32}$, S-$Zn_{30}Fe_2S_{32}$ and S-$Zn_{28}Fe_4S_{32}$ are negatively correlated with the increase of the wavelength, whereas the reflection coefficients of S-$Zn_{26}Fe_6S_{32}$ and S-$Zn_{24}Fe_8S_{32}$ are negatively correlated with the wavelengths of 400-574 nm, and positively correlated with the wavelengths of 574-760 nm. The inflection point of S-$Zn_{26}Fe_6S_{32}$ is about 635 nm (actual calculated value 634.66 nm), and the reflection coefficient is about 0.25 (actual calculated value 0.2546); the inflection point of S-$Zn_{24}Fe_8S_{32}$ is about 597 nm (actual calculated value 596.54 nm), and the reflection coefficient is 0.25 (actual calculated value 0.2518). When the solvation plane is (110) crystal plane, the reflection coefficients of S-$Zn_{32}S_{32}$ and S-$Zn_{30}Fe_2S_{32}$ are negatively correlated with the increase of wavelength, and S-$Zn_{28}Fe_4S_{32}$ enters into a flat trend after the wavelength of 678 nm and the reflection coefficient of 0.24, and the reflection coefficients of S-$Zn_{26}Fe_6S_{32}$ and S-$Zn_{24}Fe_8S_{32}$ at the wavelengths of 400-520 nm are S-$Zn_{26}Fe_6S_{32}$ has a negative correlation at 400-520 nm and a positive correlation at 520-760 nm, while S-$Zn_{26}Fe_6S_{32}$ has an inflection wavelength of about 529 nm (the actual calculated value is 529.34 nm) and a reflection coefficient of about 0.25 (the actual calculated value is 0.2532), and S-$Zn_{24}Fe_8S_{32}$ has an inflection wavelength of about 546 nm (the actual calculated value is 545.64 nm) and a reflection coefficient of 0.24 (actual calculated value 0.2372) (Fig. 7a, b).

Calculating the interstitial doping case, regardless of the solvation surface is (100) or (110) crystal surface, Fe into the lattice interstitial compared to the isomorphous substitution change situation is not obvious, but still in the red-brown band S-$Zn_{26}Fe_6S_{32}$ and S-$Zn_{24}Fe_8S_{32}$ still have a tendency to change (Fig. 7c, d).

(2)Calculation results of absorption coefficients for different doping conditions.

Calculating the substitutional doping case, when the solvation plane is (100) crystal plane, the absorption coefficients of S-$Zn_{32}S_{32}$, S-$Zn_{30}Fe_2S_{32}$, S-$Zn_{28}Fe_4S_{32}$, S-$Zn_{26}Fe_6S_{32}$, and S-$Zn_{24}Fe_8S_{32}$ show a negative correlation with the increase of wavelength. When the solvation plane is (110) crystal plane, the absorption coefficients of S-$Zn_{32}S_{32}$, S-$Zn_{30}Fe_2S_{32}$, S-$Zn_{28}Fe_4S_{32}$, and S-$Zn_{26}Fe_6S_{32}$ are negatively correlated with the increase of the wavelength, and S-$Zn_{24}Fe_8S_{32}$ is negatively correlated with the increase of the wavelength after the wavelength of 579 nm (the actual value of the calculation of 578.55 nm) and the absorption coefficient of 29712 (the actual calculation of 29712.04). 29712.04) showed positive correlation (Fig. 8a, b).

Calculating the interstitial doping case, regardless of the solvation surface is (100) or (110) crystal surface, Fe into the lattice interstitial compared to the isomorphous substitution change situation is not obvious, the trend of change is negatively correlated (Fig. 8c, d).

5. Discussion

5.1 Fe–Zn substitution pattern and its effect on the color of sphalerite

Fe easily compounds with S to form sulfides. The Fe atoms lose their outermost electrons to become ions, most of which have the same valence electrons or similar ionic radii. Since the ionic radius of Fe^{2+} is 76 ppm and that of Zn^{2+} is 74 ppm, the two ions have very close ionic radii, making Fe^{2+} the most capable of substituting for Zn^{2+}.

Previous researchers have classified the substitution mechanisms in sphalerite primarily based on ion radius into two categories: The first category involves the simple substitution of Zn^{2+} in the sphalerite lattice by a single ion, i.e., the simple replacement form of $Zn^{2+} \leftrightarrow M^{2+}$($Fe^{2+}$, Mn^{2+}, Cd^{2+}); the second category involves the substitution of Zn^{2+} in the sphalerite lattice by two or more ions, i.e., $2Zn^{2+} \leftrightarrow 2Ag^{+}/Cu^{+}+M^{2+}$($Fe^{2+}$, Mn^{2+}, Cd^{2+}), $4Zn^{2+} \leftrightarrow 2Ag^{+}/Cu^{+}+M^{2+}+M^{4+}$, and so on.

In the early mineralization stage (dark color), sphalerite is relatively depleted in Cd and enriched in Fe, while in the late mineralization stage (light color), sphalerite is relatively enriched in Cd and depleted in Fe (Li and Zhu, 2020). In the light-colored sphalerite formed during the late low-temperature stage, the geochemical parameters and substitution behavior of Cd and Fe are similar. They enter the sphalerite lattice together through isomorphous substitution of Zn, resulting in a "positive correlation" between Cd and Fe (Zhang, 2016), occupying the lattice positions that should be occupied by Fe. This, in turn, leads to the lightest color of sphalerite in the late stage.

In addition, there are substitution mechanisms involving lattice vacancies in sphalerite, which can reflect the genesis of ore deposits and provide geochemical evidence (Johan, 1988; Cook et al., 2009; Belissont et al., 2014; Ye et al., 2011; Wu et al., 2019). Studies have shown that Fe and Zn in

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sphalerite exhibit a significant negative correlation, with Zn and Fe in sphalerite displaying an isomorphic substitution relationship. The results of LA-ICP-MS Mapping analysis are consistent with those of trace element spot analysis, showing that as Fe content increases significantly, Zn content decreases, and the color of sphalerite darkens (Fig. 9). The concentration of Fe gradually decreases from black to red-brown to yellow and finally to white areas (Fig. 5).

5.2 The influence of Fe on the color of sphalerite through substitutional and interstitial doping

The reflectance (R) of a mineral describes the ratio of the energy reflected at the interface with the previous medium to the total energy when light or other waves propagate from one medium to another. It is a dimensionless value ranging from 0 to 1, with values closer to 1 indicating stronger reflection and values closer to 0 indicating weaker reflection. The absorption coefficient (α) is a physical quantity used to describe the ability of a material to absorb light of a specific wavelength. It represents the relative attenuation of light intensity per unit path length and is also a dimensionless physical quantity. Evidently, there are differences in physical meaning and calculation methods between the two, and both are dimensionless values. Therefore, these two parameters cannot be unified under a single dimension. They independently describe different ways in which materials interact with light: the reflectance coefficient describes the reflection behavior of light at the interface, while the absorption coefficient describes the attenuation behavior of light as it propagates within the material. Due to the differences in physical meaning and calculation methods between the reflectance coefficient and the absorption coefficient, and the fact that both are dimensionless values, these two parameters cannot be unified under a single dimension when calculating optical properties. They independently describe different ways in which materials interact with light: the reflectance coefficient describes the reflection behavior of light at the interface, while the absorption coefficient describes the attenuation behavior of light as it propagates within the material.

According to the DFT results, Fe enters the (100) and (110) cleavage planes in the form of isomorphic substitution and interstitial atoms. In the visible light spectrum, especially within the red-brown band, the presence of Fe in sphalerite, whether as isomorphic substitution or interstitial impurities, leads to an enhancement of reflection efficiency in this band (Fig. 7). For visual observation, a higher Fe content corresponds to a higher reflectance, resulting in a sensory enhancement of color within this band. In terms of atomic existence forms, as the Fe content increases, the reflectance coefficient of Fe present as interstitial impurities in sphalerite is higher than that of Fe present as isomorphic substitutions. As for the cleavage surfaces, with increasing Fe content, regardless of the form in which it enters the sphalerite lattice, the reflectance coefficient of the (110) plane is slightly higher than that of the (100) plane. As the Fe content increases, the absorption coefficient also shows an increasing trend in the visible light spectrum, particularly within the red-brown band (Fig. 8). In terms of atomic existence forms, the absorption coefficient of isomorphic substitutions is higher than that of interstitial atoms, and the absorption coefficient of the (110) plane is slightly higher than that of the (100) plane.

It can be seen that in the red-brown band, especially when Fe atoms are present in the form of isomorphic substitution, the reflectance coefficient in this band is significantly higher than that in other bands within the visible light range when there are more than six Fe atoms, while the absorption coefficient is lower than that in other bands.

In summary, when Fe enters the sphalerite lattice in the form of isomorphic substitution, as its content increases, the reflectance coefficient in the red-brown band enhances, and the absorption coefficient in the non-red-brown range also increases, resulting in a more pronounced red-brown color under visual observation. This phenomenon is particularly evident on the (110) cleavage plane when Fe is present as isomorphic substitution.

5.3 The coloration mechanism of sphalerite and its relationship with the physicochemical conditions of mineralization

Through LA-ICP-MS data analysis, the coloration mechanism of Fe in sphalerite is primarily based on the substitution of Fe for Zn. When the degree of Fe substitution for Zn is low, Zn and S are connected by covalent bonds, and the mineral exhibits the optical properties of an atomic lattice crystal. In this case, sphalerite may display relatively light colors such as white or yellow. As the amount of Fe substitution increases, and when the degree of Fe substitution for Zn is high, the chemical bonds between Fe and S become covalent, leading to the mineral exhibiting certain metallic optical properties. Consequently, the color gradually darkens, appearing as red-brown, black, etc.

Based on DFT calculations, the changes in the electronic structure of sphalerite after Fe substitution can be revealed. When Fe substitutes for Zn in sphalerite, it introduces impurity energy levels within the forbidden band of sphalerite, which are composed of Fe's 3d orbitals. These impurity energy levels affect the light absorption and reflection characteristics of sphalerite, thereby altering its color. In sphalerite (ZnS), Fe atoms can substitute for Zn atoms, forming isomorphic mixed crystals. This substitution alters the chemical composition and lattice structure of sphalerite, further influencing physical processes such as light absorption and reflection, ultimately leading to color changes. Sphalerite with lower Fe content may exhibit lighter colors, while sphalerite with higher Fe content may display darker colors. The crystal planes of sphalerite influence the absorption and reflection of light on its surface, with different crystal planes possibly having stronger absorption or reflection effects on specific wavelengths of light, resulting in color differences. The fact that Fe mainly enters the (110) cleavage plane in the form of isomorphic substitution and interstitial atoms demonstrates the high sensitivity of the (110) cleavage plane to Fe substitution, making it a contributing factor to the different colors of sphalerite.

Studies have shown that trace elements in sphalerite can effectively indicate mineralization temperatures (Li et al., 2022; Vincent et al., 2021; Zhu et al., 2022; Hu et al., 2023). The Fe content in sphalerite is controlled by temperature (Frenzel et al., 2016; Liu et al., 2020). Niu et al. (2023) constructed two sequencing models for sphalerite, in which the different colors of sphalerite indicate that as the Fe content gradually decreases, the temperature also gradually decreases, suggesting a clear positive correlation between temperature and Fe content (Niu et al., 2023; Lai X, 2023). The mineralization temperature gradually decreases from the early stage to the late stage. During the mineralization process, the pH value, oxygen fugacity (fO_2), sulfur fugacity (fS_2), and salinity of the ore-forming fluid also jointly affect the precipitation and color changes of sphalerite. In the migration stage, the fluid is acidic, but as mineralization progresses, the pH value gradually increases, reaching near-neutral to weakly alkaline conditions in the sphalerite-galena stage, promoting the massive precipitation of sphalerite and galena. The color of sphalerite gradually becomes lighter as the pH value increases, with darker colors formed earlier. In the oxygen fugacity-sulfur fugacity diagram, the minimum $logfO_2$ and $logfS_2$ required for the formation of sphalerite are the lowest. Changes in oxygen fugacity also control the precipitation of sphalerite, which requires a relatively low oxygen fugacity. In the sphalerite-galena stage, as the oxygen fugacity increases, sphalerite precipitates first, followed by galena, and the color of sphalerite also becomes lighter with changes in oxygen fugacity. Lower fS_2 conditions are favorable for the precipitation of sphalerite, and at this time, the color of sphalerite is lighter. As fS_2 increases, the color may darken (Fig. 10). The salinity of fluid inclusions in sphalerite varies greatly, and medium-to-high salinity environments are conducive to the precipitation of sphalerite, with a decrease in salinity accompanying a lightening of the color of sphalerite. In summary, the precipitation and color changes of sphalerite are influenced by multiple factors working together (Zhang et al., 2014, 2017, 2023).

As mentioned earlier, the first stage of sphalerite in the Huize Pb-Zn ore is black-reddish brown, the second stage of sphalerite is reddish brown, and the third stage of sphalerite is yellow and white. In the initial stage of mineralization, the hydrothermal fluid temperature is relatively high and the pH value is low (Yan Zhang et al., 2023); the metal ion activity and sulfhydryl ion activity required for the precipitation of pyrite and sphalerite under near-neutral conditions are very similar, but they do not reach the logfO_2, logfS_2 required for the precipitation of pyrite, while the radii of Fe^{2+} and Zn^{2+} are very close to each other, so that homogeneous-like substitutions can occur, and a large amount of Fe^{2+} in the fluid is easier to enter into the sphalerite. A large number of Fe^{2+} is easier to enter the sphalerite lattice, so the sphalerite formed at this stage is darker in color, black-reddish brown. When the mineralization enters the second stage, the temperature begins to decrease, logfO_2 gradually increases, although logfS_2 decreases, but the decrease in temperature makes the decrease in logfS_2 required for mineral precipitation even greater (Zhang Yan et al., 2023), and the amount of Fe^{2+} in the fluid that enters the sphalerite lattice decreases, so that sphalerite at this stage becomes lighter reddish-brown in color. In the third stage of mineralization, the mineralization temperature continued to decrease, and the Fe^{2+} content in the fluid into the sphalerite lattice further decreased; coupled with the large amount of precipitation of Zn in the previous stage, so that its concentration in the fluid decreased, and with the upward and outward transport of the fluid, logfO_2 increased, so that Zn is easy to be transported in the form of ions with the fluid (Yan Zhang et al., 2023), and the content of sphalerite in this stage decreased, and its color is lighter, and it is yellow and white. The color of sphalerite is yellow and white.

In summary, during the formation of sphalerite, external factors such as temperature, pressure, pH, fO_2, and fS_2 influence the migration and enrichment of ore-forming fluids, indirectly affecting the internal factors of sphalerite formation, including its chemical composition, physical properties, and crystal structure. In the microscopic Fe mainly enters into the solution surface of sphalerite (110) in the form of homogeneous-like and interstitial atoms, isomorphic substitution is that Zn atoms in sphalerite are directly replaced by Fe atoms, while interstitial substitution is that Fe atoms first exist in an interstitial state with gaps between atoms inside the crystal, and then Zn atoms are kicked out to occupy the original positions in the crystal lattice. Both substitution methods have some effect on the nature of the crystal, but this effect is usually small, transitional, will not cause qualitative changes in the crystal structure and bonding, but the analogous isomorphic substitution due to Fe atoms directly replace the lattice atoms (Zn), and therefore have a significant effect on the color of sphalerite, along with the reduction of the Fe content, and changes in the physicochemical conditions of the formation of sphalerite, the color of the sphalerite by the black−red-brown With the reduction of Fe content and changes in the physicochemical conditions of mineralization, the color of sphalerite changes from black to red-brown to yellow to white, corresponding to different stages of mineralization. By affecting the Fe^{2+} content in the sphalerite lattice, the color of sphalerite becomes lighter with the progress of mineralization, which further supports the fact that the metallogenic fluids in the Huize Pb−Zn deposits show a medium-high-temperature−low-salinity → medium-temperature−moderate-salinity → low-temperature-low−moderate-salinity evolutionary process (Han et al., 2020).

6. Conclusions

(1) In the Huize Pb–Zn deposit, the higher the Fe content in sphalerite, the darker its color, with the black sphalerite from the Baizuo Formation having the highest Fe content and the red-brown sphalerite from the Dengying Formation also having the highest Fe content among its kind.

(2) When the number of Fe atoms in the ZnS electronic structure exceeds six, the reflection coefficient in the red-brown spectral range is significantly higher than that in other visible light bands, while the absorption coefficient is lower. The isomorphous substitution and interstitial insertion of Fe in sphalerite are key factors leading to its color variation. Fe predominantly enters the (110) and (100) cleavage planes in the form of isomorphous substitution, forming isomorphous mixed crystals, with the (110) cleavage plane being more sensitive to Fe substitution.

(3) There is a positive correlation between mineralization temperature and Fe content in sphalerite, which determines the color presentation of sphalerite. From the early to the late mineralization stages in the Huize Pb–Zn deposit, the color of sphalerite changes from dark (black, red-brown) to light (yellow, white), reflecting a decrease in mineralization temperature and an evolution of the ore-forming fluid from medium-to-high temperature, and low salinity to medium temperature, and medium salinity, and finally to low temperature, and medium-to-low salinity.

Data availability

All data used in this study are available in the Table/Supplementary Information/Data files and published literature.

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Process and Mineral Resources Innovation Central (2012).

Declaration of interest statement

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References

Adachi, S., 2012. Optical properties of crystalline and amorphous semiconductors: Materials and fundamental principles. *Springer Science & Business Media*.

Belissont, R., Boiron, M.C., Luais, B., and Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac–Saint-Salvy deposit (France): Insights into incorporation mechanisms and ore deposition processes. *Geochimica et Cosmochimica Acta*, v. 126, p. 518–540.

Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Danyushevsky, L., Shimizu, M., SainiEidukat, B., and Melcher, F., 2009. Trace and minor elements in sphalerite: A LA-ICP-MS study. *Geochimica et Cosmochimica Acta*, v. 73, p. 4761–4791.

Danyushevsky, L., Robinson, P., Gilbert, S., Norman, M., Large, R., McGoldrick, P., and Shelley, M., 2011. Routine quantitative multi-element analysis of sulphide minerals by laser ablation ICP-MS: Standard development and consideration of matrix effects. *Geochemistry*, v. 11, p. 51–60.

Frenzel, M., Hirsch, T., and Gutzmer, J., 2016. Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type-a meta-analysis. *Ore Geology Reviews*, v. 76, p. 52–78.

Han, R.S., Hu, Y.Z., Wang, X.K., Hou, B.H., Huang, Z.L., Chen, J., Wang, F., Wu, P., Li, B., Wang, H.J., Dong, Y., and Lei, L., 2012. Mineralization model

Journal Pre-proofs

of rich Ge–Ag–bearing Zn–Pb polymetallic deposit concentrated district in Northeastern Yunnan, China. *Acta Geologica Sinica*, v. 86, p. 280–293.

Han, R.S., Wu, J.B., Zhang, Y., Chen, Q., and Sun, B.T., 2024. Oblique distribution patterns and the underlying mechanical model of orebody groups controlled by structures at different scales, *Scientific Reports*, v. 14, p. 4591.

Han, R.S., Wu, P., Wang, F., Zhou, G.M., Li, W.Y., and Qiu, W.L., 2019. Four Steps Type" ore–prospecting method for deeply concealed hydrothermal ore deposits—A case study of the Maoping Zn–Pb (Ag–Ge) deposit in southwestern China. *Geotectonica et Metallogenia*, v. 43, p. 246–257.

Han, R.S., Wu, P., Zhang, Y., Huang, Z.L., Wang, F., Zhou, G.M., Shi, Z.L., and Zhang, C.Q., 2022, New progresses of metallogenic theory for rich Zn–Pb–(Ag–Ge) deposits in the Sichuan–Yunnan–Guizhou Triangle (SYGT) area, Southwestern Tethys. *Acta Geologica Sinica*, v. 96, p. 554–573.

Han, R.S., Xue, C.D., Hu, Y.Z., and Huang, Z.L., 2006. Enrichment features and significances of Ag and dispersed elements in the ores in the Huize Carbonate-hosted Zn–Pb–(Ag–Ge) district, Yunnan, China. *Geochimica et Cosmochimica Acta*, v. 70, p. A544.

Han, R.S., Zhang, Y., Ye, T.Z., Chen, Q., Ren, T., Guo, Z.L., and Qiu, W.L., 2023, An Overview of the Metallogeny and Geological Prospecting Model of Mississippi Valley Type (MVT) Lead and Zinc Deposits. *Geotectonica et Metallogenia*, v. 47, p. 915–932.

Hu, R.Z., Fu, S., Huang, Y., Zhou, M.F., Fu, S., Zhao, C., Wang, Y., Bi, X. and Xiao, J.,2017. The giant South China Mesozoiclow-temperature metallogenic domain: reviews and a new geodynamic model. *Journal of Asian Earth Sciences*, v. 137, p. 9–34.

Hu, X.J., Li, H., Forster, M.W., Elatikpo, S.M., Kong, H., Wu, J.H., and Zhu, D.P., 2023. Using apatite to differentiate metallogenic potential and environment of granitic rocks: A case study from the Tongshanling W–Sn–Cu–Pb–Zn ore field, Nanling Range (South China). *Journal of Geochemical Exploration*, v. 245, p. 107143.

Journal Pre-proofs

Jia, F.D., Zhang, C.Q., Hua, Z.X., Lou, D.B., Li, B.L and Sun, J., 2020. Identification characteristics, composition and genesis of euclase in Mahuaping tungsten-beryllium polymetallic deposit in Yunnan Province, southwest China. *Spectroscopy and Spectral Analysis*, v. 40, p. 3185–3192.

Johan, Z., 1988. Indium and germanium in the structure of sphalerite: an example of coupled substitution with copper. *Mineralogy and Petrology*, v. 39, p. 211–229.

Kresse, G., & Furthmüller, J., 1996. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. Computational Materials Science, v. , p. 15-50.

Lai, X., Chen, C.H., Yang, Y.L., Liu, S.Y., Li, Y., Wang, J.X., Song, Z.J., Gu, Y., Chen, X.J. and Huang, X.D., 2023. Constraints on Metallogenic Temperature and Mineralization Style from Reflection Color of Sphalerite. *Ore Geology Reviews*, v. 161, p. 105634.

Li, L.J., Han, R.S., Zhang, Y., Wu, J.B., and Feng, Z.X., (2022). Trace element signatures of sphalerite in the Sichuan Daliangzi Ge-rich Pb–Zn deposit and its implications for deep ore prospecting. *Frontiers in Earth Science*, v. 10, p. 928738.

Li, X.D., 2022. Vertical variation of physical and chemical conditions of ore-forming fluids in Maoping. Kunming University of Science and Technology.

Li, Y.G., and Zhu, C.W., 2020. The distribution signatures of major and trace elements in zoned sphaleritefrom lead–zine deposits: A case study from the Huize deposit. *Acta Mineralogica Sinica*, v. 40, p. 765–771.

Li, Z.L., Ye, L., Hu, Y., Wei, C., Huang, Z.L., Yang, Y.L., and Danyushevsky, L., 2020. Trace elements in sulfides from the Maozu Pb–Zn deposit, Yunnan Province, China: Implications for trace-element incorporation mechanisms and ore genesis. *American Mineralogist*, v. 10, p. 1734–1751.

Liu, H., Zhang, C.Q., Ji, X.J., Guo, Z.L., Lou, D.B., Wu, Y., Zhang Y.F., and Li, Y.L. , 2022. Differential enrichment of germanium in sphalerite from Huize lead–zinc deposit. *Yunnan Province Miner Deposits*, v. 41, p. 1057–1072.

Journal Pre-proofs

Liu, H.C. and Lin, W.D.,1999. A Study on the Regularity of Pb–Zn Silver Deposits in Northeast Yunnan 40–76 (Yunnan University Press, 1999).

Liu, J.B., Zhang, D.X., Hu, Z.Q., Chen, S.W., and Xie, X.Y., 2024. Mineralogy, Trace Element Chemistry, and Their Metallogenic Implications of Sphalerites from the Haopinggou Ag–Au–Pb–Zn Polymetallic Deposit, Xiong'ershan Area, Western Henan. *Geoscience*, v. 38, p. 198–213.

Liu, T.T., Zhu, C.W., Yang, G.S., Zhang, G.S., Fan, H.F., Zhang, Y.X., and Wen, H.J., 2020. Primary study of germanium isotope composition in sphalerite from the Fule Pb–Zn deposit, Yunnan province. *Ore Geology Reviews*, v. 120, p. 103466.

Liu, Y.J., Zhang, J.R., Qiao, E.G., Mao, H.X., and Zhao, M.F., 1984. Geochemistry of gold deposits in west Hunan and East Guangxi, China. *Geochemistry*, v. 3, p. 307–321.

Liu, Y.D., Ang, W., An, P., Li, Y. L., Kong, Z.G., Zhang, C.Q., Li, F., and Wang, Y., 2024. Geochemical Characteristics and Indication Significance of Jinshachang Lead–zinc Deposit, Yunnan Province. *Acta Geoscientica Sinica*, v. 45, p. 530–546.

Liu, Y.S., Hu, Z.C., Gao, S., Günther, D., Xu, J., Gao, C.G., and Chen, H.H.,2008. In situ analysis of major and trace elements of anhydrous minerals by LA-ICP-MS without applying an internal standard. *Chemical Geology*, v. 257, p. 34–43.

Lucarini, V., Saarinen, J. J., Peiponen, K. E., & Vartiainen, E. M.,2005. Kramers-Kronig relations in optical materials research. Springer *Science & Business Media*, v. 110.

Martin, R. M., 2004. Electronic Structure: Basic Theory and Practical Methods. *Cambridge University Press*.

Nassau, K., 1978. The origins of color in minerals. *American mineralogist*, v. 63, p. 219–229.

Niu, P.P., Muñoz, M., Mathon, O., Xiong, S.F., and Jiang, S.Y., 2024. Mechanism of germanium enrichment in the world-class Huize MVT Pb–Zn deposit, southwestern China. *Mineralium Deposita*, v. *59*, p. 995–1016.

Paton, C., Hellstrom, J., Paul, B., Woodhead, J., and Hergt, J., 2011. Iolite: Freeware for the visualisation and processing of mass spectrometric data. *Journal of Analytical Atomic Spectrometry*, v. 26, p. 2508–2518.

Qiu, L., Yan, D.P ., Tang, S.L., Wang, Q., Yang, W.X., Tang, X., and Wang, J., 2016. Mesozoic geology of southwestern China: indosinian foreland overthrusting and subsequent deformation. *Journal of Asian Earth Sciences*, v. 122, p. 91–105.

Tran, F., & Blaha, P. (2009). Bandgap calculations and trends of ordered alloys, ordered vacancies, and antisite defects in CuInSe2 and CuGaSe2. *Journal of Physics: Condensed Matter*, v. 21, p. 395502.

Vincent, V.I., Li, H., Girei, M.B., Ahmed, H.A., and Ntekim, E.E., 2021. Genesis and age of Pb−Zn mineralization from the Ningi-Burra ring complex, North Central Nigeria: Constraints from zircon morphology, U−Pb dating and Lu−Hf isotopes. *Lithos*, v. 390, p. 106115.

Wang, B.L., Li, L.H., and Zeng, P.S., 2004. Basic geophysic characteristics of Sichuan–Yunnan–Guizhou rhombic block and its relationship with endo-mineralization. *Journal of East China Institute of Technology*, v. 27, p. 301–308.

Wang, J.Y., 2020. Study on the Gemological and Mineralogical Characteristics of Colored Sphalerite. China University of Geosciences(Beijing).

Wang, Q., Gu, X.X., Fu, S.H., Zhang, M., and Li, F.Y., 2008. Enrichment of the dispersed elements Cd, Ge and Ga in the Huize lead–zinc deposit, Yunnan. *Sedimentary Geology and Tethyan Geology*, v. 28(04):69–73.

Wang, Z.Q.,2017. Enrichment Regularity of Dispersed Elements in Huize Super-Large Pb–Zn Deposit. China University of Geosciences(Beijing).

Wei, C., Ye, L., Danyushevskiy, L., Li, Z.L., Hu, Y.S., and Huang, Z.L., 2019. Distribution and occurrence of Ge and related trace elements in sphalerite from the Lehong carbonate-hosted Zn–Pb deposit, northeastern Yunnan, China: Insights from SEM and LA-ICP-MS studies. *Ore Geology Reviews*, v. 115, p. 103–175.

Wilson, S.A., Ridley, W.I., and Koenig, A.E., 2002. Development of sulfide

Journal Pre-proofs

calibration standards for the laser ablation inductively-coupled plasma mass spectrometry technique. *Journal of Analytical Atomic Spectroscopy*, v. 17, p. 406–409.

Wu, M. Q., Samson, I. M., Zhang, D. H., 2017. Textural and chemical constraints on the formation of disseminated granite-hosted W-Ta-Nb mineralization at the Dajishan Deposit, Nanling Range, Southeastern China. *Economic Geology*, v.112, p. 855–887.

Wu, J.B., Han, R.S., Zhang, Y., Sun, B.T, Li, W.Y., Li, D.Q, Cao, Y.R., and Cen, C.,2024a. The fault–fold structure ore control mechanism of hydrothermal deposits——A case study of the Maoping super-large rich-Ge lead–zinc deposit in northeastern Yunnan, China. *Ore Geology Reviews*, v.168, p.106039.

Wu, J.B., Han, R.S., Zhang, Y., Wu, P., Gong, H.S.,Wang, L., Cheng, G.,Li, X.D.,Yang, Y.X. and Mi, Y.Y.,2024b. Porosity–permeability characteristics and mineralization–alteration zones of the Maoping germanium-rich lead–zinc deposit in SW China. *Frontiers in Earth Science*, v. 12, p. 1347243.

Wu, J.B., Pi, Q.H., Zhu, B., Hu, Y.Y., Li, G., and Wei, C.W., 2020. Late Cretaceous-Cenozoic exhumation of Northwestern Guangxi (China) and tectonic implications: evidence from apatite fission track dating. *Geochemistry*, v. 80, p. 125662.

Wu, Y., Kong, Z.G., Chen, M.H., Zhang, C.Q., Cao, L., Tang, Y.J., Yuan, X.,and Zhang, P., 2019. Trace elements in sphalerites from the Mississippi Valley-type lead–zinc deposits around the margins of Yangtze Block and its geological implications: A LA-ICP-MS study. *Acta Petrologica Sinica*, v. 35, p. 3443–3460.

Wu, Z., Tang, X., Feng, Y., Li, X., and Chen, Y., 2024. Effect of natural non-metallic impurities on the electronic structure and optical properties of sphalerite ZnS. *Micro and Nanostructures*, v. *189*, p. 207818.

Xing, B., Mao, J.W., Xiao, X.N., Liu, H., Zhang, C., Guo, S., Li, H.Y., Huang, W.Y., and Lai, C., 2022. Metallogenic discrimination by sphalerite trace element geochemistry: An example from the Fengyan Zn–Pb deposit in central Fujian, SE China-ScienceDirect. *Ore Geology Review*, v. 141, p.

Journal Pre-proofs

104651.

Xiong, X.F., 2013. The Geochemical characteristics of Fozichong Pb–Zn Ore Field, Guangxi. Yangtze University.

Yan, D.P., Zhou, M.F., Song, H.L., Wang, X.W., and Malpas, J., 2003.Origin and tectonic significance of a mesozoic multi-layer over-thrust system within the Y angtze block (South China). *Tectonophysics*, v. 361, p. 239–254.

Yang, Q., Zhang, X.J., Ulrich, T., Zhang, J., and Wang, J., 2022. Trace element compositions of sulfides from Pb–Zn deposits in the Northeast Yunnan and northwest Guizhou Provinces, SW China: Insights from LA-ICP-MS analyses of sphalerite and pyrite. *Ore Geology Reviews*, v. *141*, p. 104639.

Ye, L., Cook, N.J., Ciobanu, C.L., Liu, Y.P., Zhang, Q., Liu, T.G., Gao, W., Yang, Y.L., and Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in south china: a la-icpms study. *Ore Geology Reviews*, v. 39, p. 188–217.

Zhang, M.F., Zhou, Z.G., Chen, G.L., and Li, H., 2015. Characteristics and significance of the chemical composition of sphalerite in the Huize lead–zinc deposit, Yunnan. *Geological Review*, v. 61, p. 436–437.

Zhang, W.D., You, H.T., Li, B., Zhao, K.D., Chen, X.D., and Zhu, L., 2022. Ore-forming processes of the Qixiashan carbonate-hosted Pb–Zn deposit, South China: Constraints from sulfide trace elements and sulfur isotopes. *Ore Geology Reviews*, v. 143, p. 104786.

Zhang, Y., Han, R. S., Wu, P., Zhou, G. M., Wei, P. T., Qiu, W. L.,2014. Constraints of oxygen and sulfur fugacities on the co-existence and differentiation of lead and zinc in Huize-type lead-zinc deposits: A case study of the Zhaotong lead-zinc deposit in northeast Yunnan. *Geotectonica et Metallogenia*, V. 38, P. 898-907.

Zhang, Y., 2016. Fluid mixing metallogenic mechanism ofHuize superlarge Lead–Zinc Deposits innortheast of Yunnan poly-metallicmineralization domain. Kunming University of Science and Technology.

Zhang, Y., Han, R. S., Wei, P. T., Qiu, W. L.,2017. Characteristics of fluid inclusions and physicochemical conditions of mineralization in the Kuangshanchang lead-zinc deposit, Huize, Yunnan. *Journal of Jilin University (Earth Science Edition)*, v. 47, p. 719-733.

Zhang, Y., Chen, H., Cheng, J., Tian, J., Zhang, L., and Olin, P., 2022. Pyrite geochemistry and its implications on Au–Cu skarn metallogeny: An example from the Jiguanzui deposit, Eastern China. *American Mineralogist*, v. 107, p. 1910–1925.

Zhang, Y., Han, R.S., Ding, X., He, J.J., and Wang, Y.R., 2019. An experimental study on metal precipitation driven by fluid mixing: implications for genesis of carbonate-hosted lead–zinc ore deposits. *Acta Geochimica*, v. 38, p. 202–215.

Zhang, Y., Han, R.S., Hu, T.C., Wei, P.T., and Wang, L., 2023. The Tectonics-fluids-mineralization Coupling Processes in the Huize Ultra-large Ge-rich Pb–Zn Deposit. *Geotectonica et Metallogenia*, v. 47, p. 969–983.

Zhang, Y., Han, R.S., Wei, P.T., and Wang, L., 2017. Identification of two types of metallogenic fluids in the ultra-large Huize Pb–Zn deposit, SW China. *Geofluids*, v. 2017, p. 1–22.

Zhu, D.P., Li, H., Tamehe, L.S., Jiang, W.C., Wang, C., and Wu, K.Y., 2022. Two-stage CuPbZn mineralization of the Baoshan deposit in southern Hunan, South China: Constraints from zircon and pyrite geochronology and geochemistry. *Journal of Geochemical Exploration*, v. 241, p. 107070.

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Table 1. Description of sample characteristic

Sample	Elevation	Ore-bearing layer	Sample Descriptions
HZHK-14	1404m	*Zbdn*	The ore minerals are galena, sphalerite, and a small amount of pyrite. The galena is massive, with medium to coarse-grained sphalerite wrapped around it, and the pyrite is a fine-grained aggregate distributed at the bottom of the galena.
HZHK-15	1404m	*Zbdn*	The ore minerals are mainly pyrite, followed by galena, and sphalerite is the least abundant. The pyrite appears as a bright yellow fine-grained aggregate, the galena occurs in massive form, and the sphalerite is present as yellow-brown fine to medium-grained particles.
HZHK-39	1031m	C_1b	The ore minerals are dominated by pyrite, followed by galena, with sphalerite at least. Pyrite is a fine-grained aggregate, galena is produced in clusters next to pyrite, and sphalerite is distributed sporadically in fine-grained form.
HZHK-41	1031m	C_1b	Ore minerals are dominated by galena, followed by sphalerite, pyrite is the least, galena is in the form of lumps wrapped with yellow-brown sphalerite, sphalerite is produced in the form of black-red-brown lumps, with gray-white dolomite developed next to it, and pyrite is in the form of fine grains.
HZHK-44	1031m	C_1b	The ore minerals are dominated by galena, followed by pyrite, with sphalerite being the least, galena and pyrite in a banded structure, and sphalerite in a fine-grained form wrapped by galena.
HZHK-45	1031m	C_1b	The ore minerals are dominated by sphalerite, followed by galena, with pyrite being the least. Sphalerite is produced in red-brown masses, wrapped in lead-gray galena ore bodies, with occasional fine-grained pyrite on the surface.

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Table 2. Doping System Table

Model	Substitution doping	Model	Interstitial doping
Model S_1	$S\text{-}Zn_{32}S_{32}$	Model I_1	$I\text{-}Zn_{32}S_{32}$
Model S_2	$S\text{-}Zn_{30}Fe_2S_{32}$	Model I_2	$I\text{-}Zn_{30}Fe_2S_{32}$
Model S_3	$S\text{-}Zn_{28}Fe_4S_{32}$	Model I_3	$I\text{-}Zn_{28}Fe_4S_{32}$
Model S_4	$S\text{-}Zn_{26}Fe_6S_{32}$	Model I_4	$I\text{-}Zn_{26}Fe_6S_{32}$
Model S_4	$S\text{-}Zn_{24}Fe_8S_{32}$	Model I_4	$I\text{-}Zn_{24}Fe_8S_{32}$

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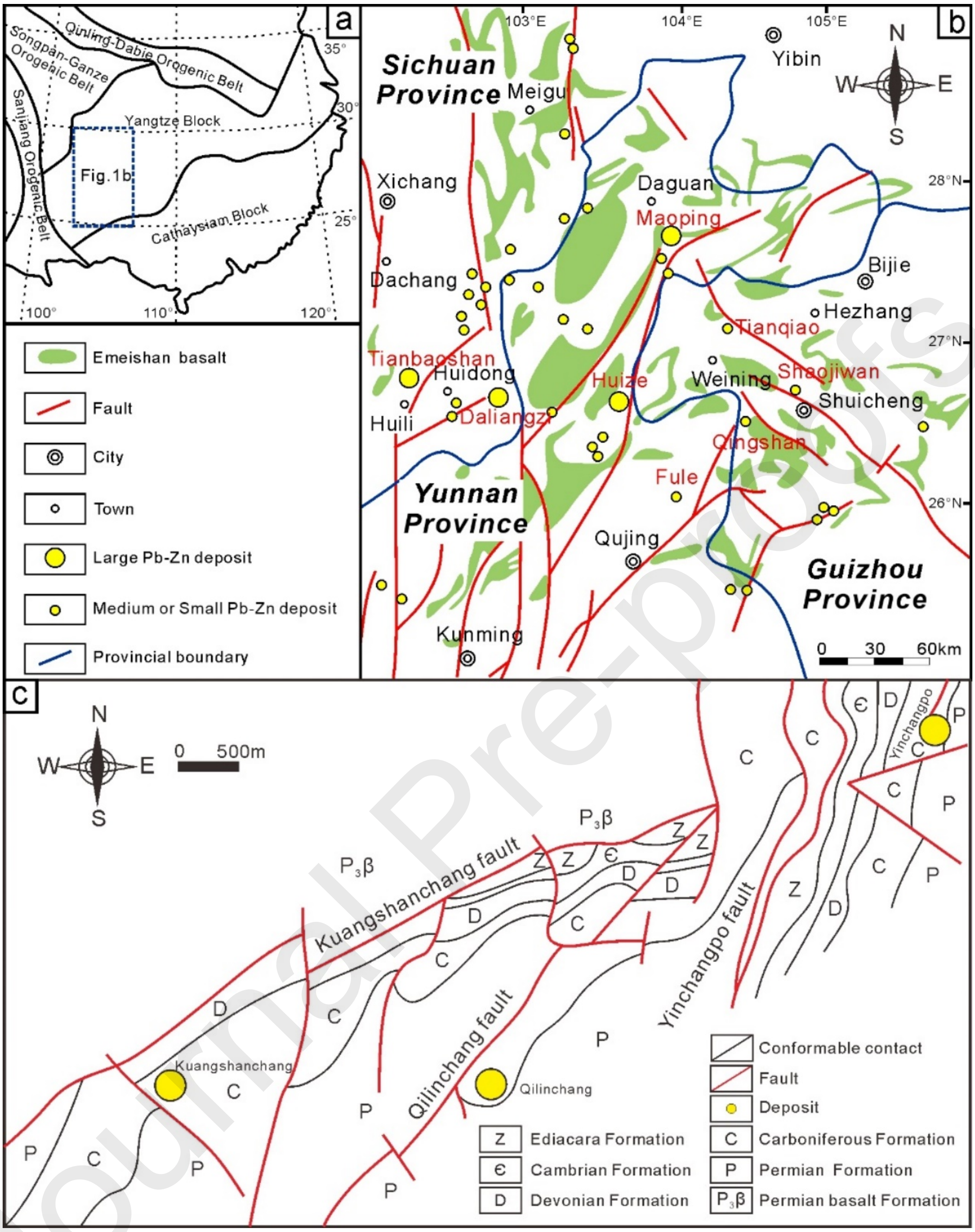

Fig. 1. a. Simplified geologic map of the South China land mass and adjacent areas (adapted from Yan et al., 2003; Qiu et al., 2016; Hu et al., 2017; Wu et al., 2020, 2024b) of the tectonic framework and the distribution of ore deposits; b. The main fault–fold tectonic zones and ore deposit distribution in the Sichuan–Yunnan–Guizhou Pb–Zn polymetallic metallogenic region (SYGD) (Han et al., (2019) redrawn according to Liu and Lin (1999)); c. Geological map of the Huize Pb–Zn deposit (Han et al., 2006).

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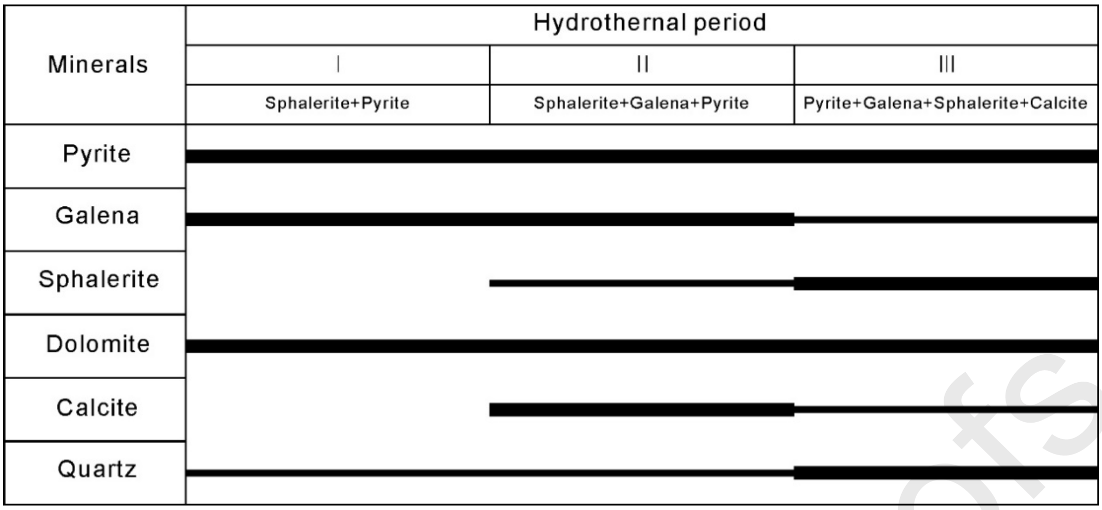

Fig. 2. Division of Pb–Zn deposit mineralization stages in Huize.

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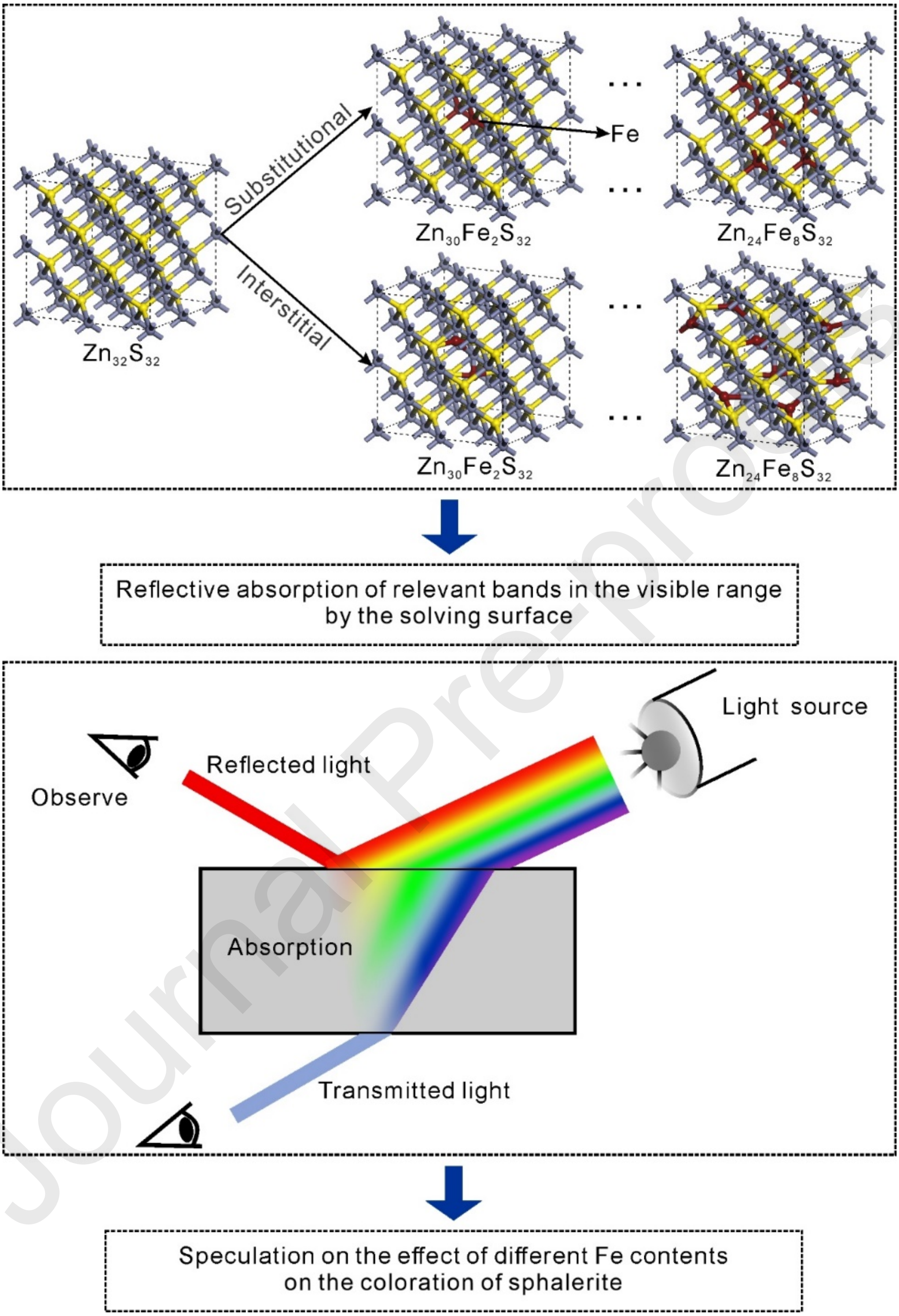

Fig. 3. Flowchart of density flood theory calculation method.

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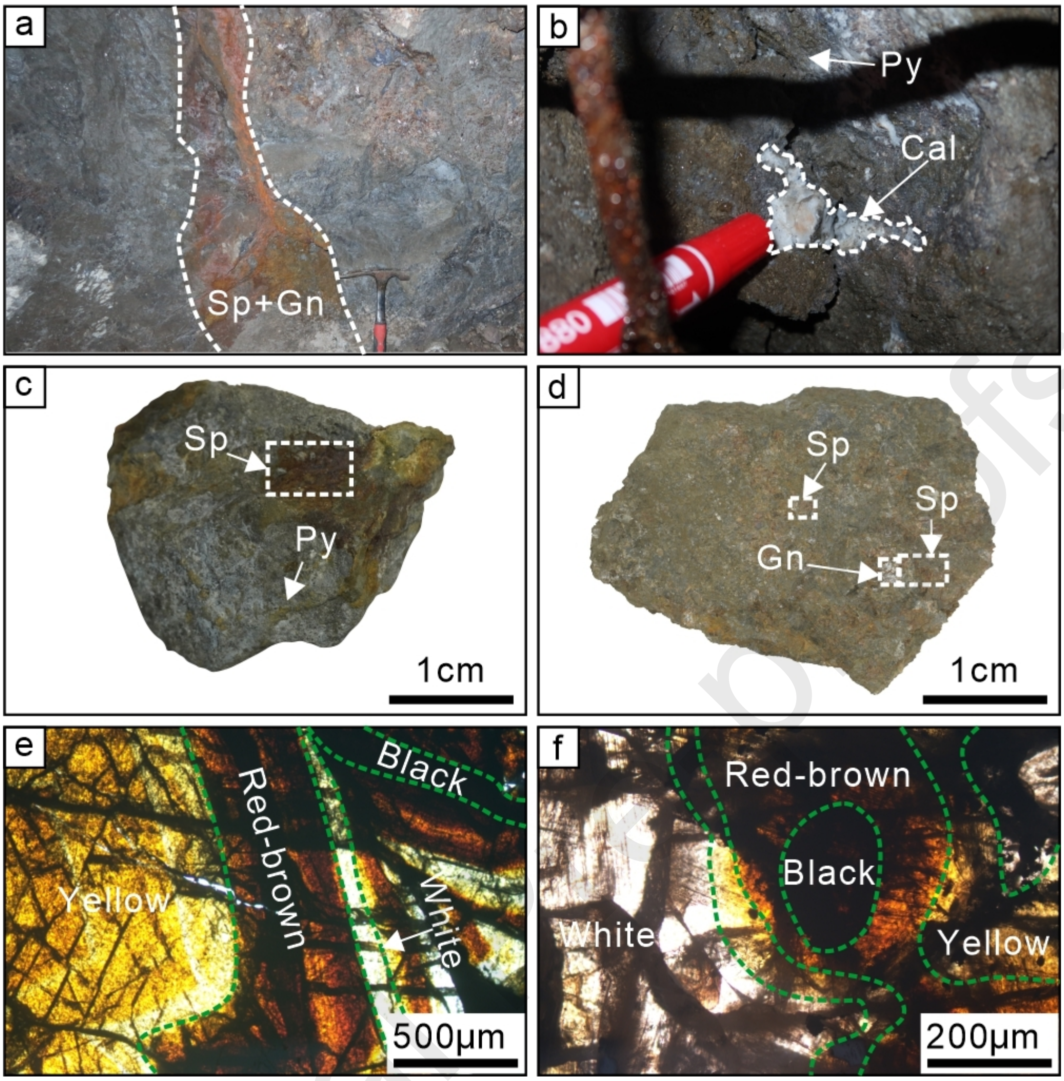

Fig. 4. Typical photo of sphalerite in the Huize lead-zinc deposit (a). The sphalerite and galena in the Dengying Formation are distributed in a vein like pattern; (b) The sphalerite and galena in the Baizuo Formation are distributed in a vein like pattern; (c) Dengying Formation blocky sphalerite and pyrite ores; (d) Baizuo Formation blocky sphalerite and galena ores; (e) Lamp shadow group sphalerite ring, belonging to jumping band, with alternating colors of the ring; (f) The banded sphalerite of the Baizuo Formation has a darker core color

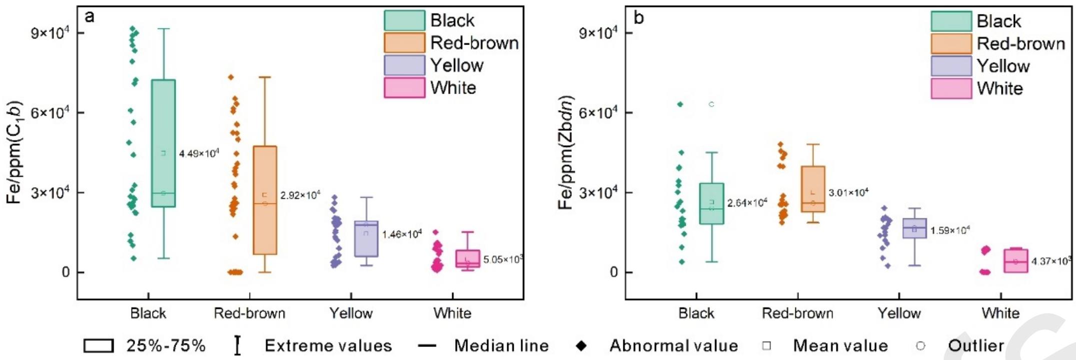

Fig. 5. Box plot of LA-ICP-MS Fe element content in sphalerite of different colors from the Baizuo Formation (a); Box plot of LA-ICP-MS Fe element content in sphalerite of different colors from the Dengying Formation (b).

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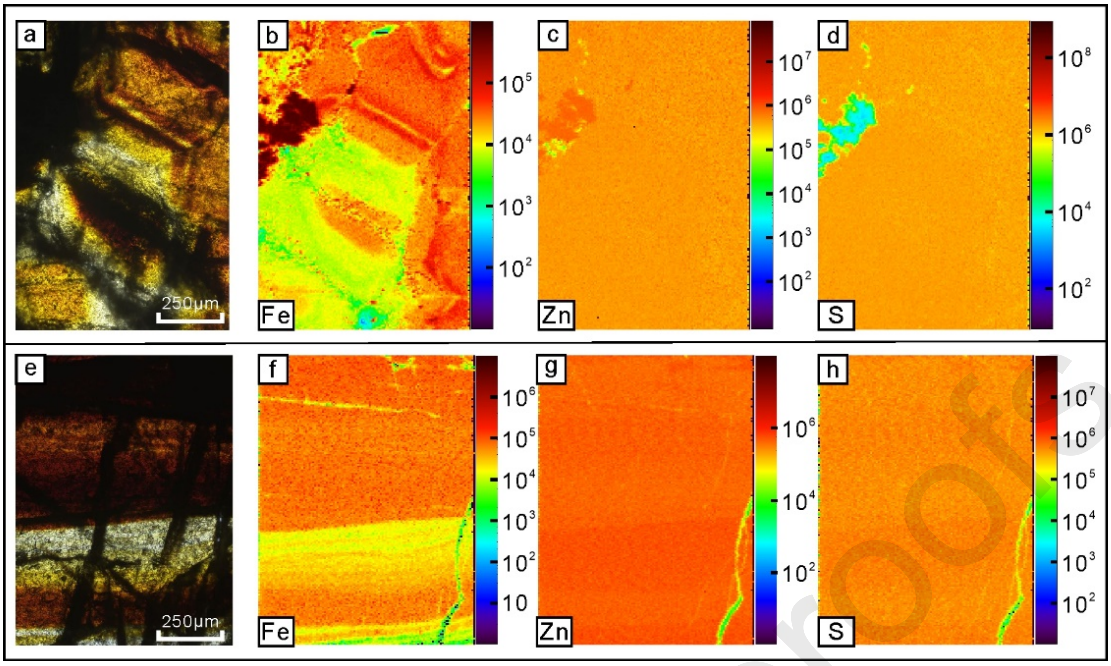

Fig. 6. LA-ICP-MS Mapping images of samples from the baizuo group (a); transmitted light micrographs of samples of ring-banded sphalerite (b-d); LA-ICP-MS Mapping images of samples from the dengying group (e); transmitted light micrographs of samples of ring-banded sphalerite (f-h).

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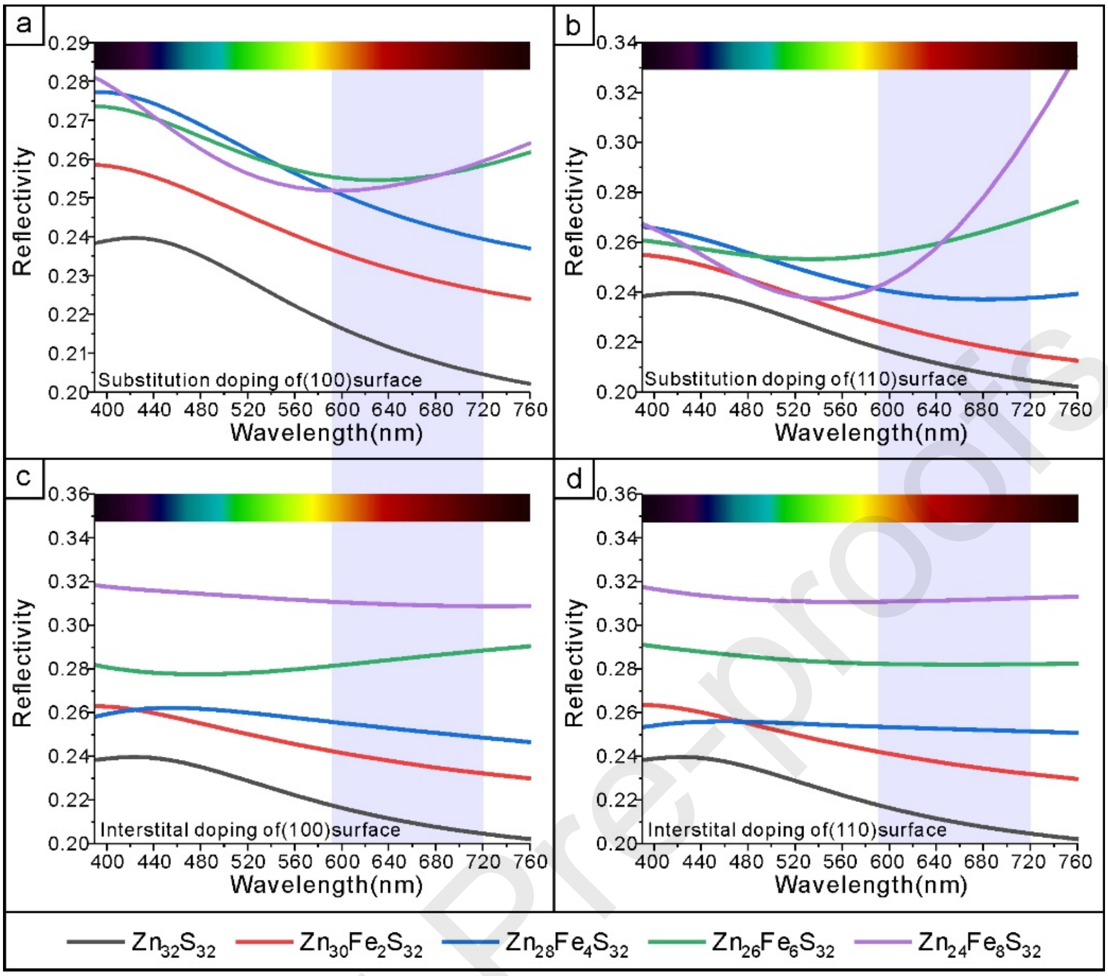

Fig. 7. Variation of reflection coefficients of Fe into the lattice interstitial solvation surfaces (100), (110) in the form of mass-like isomers and interstitial atoms.

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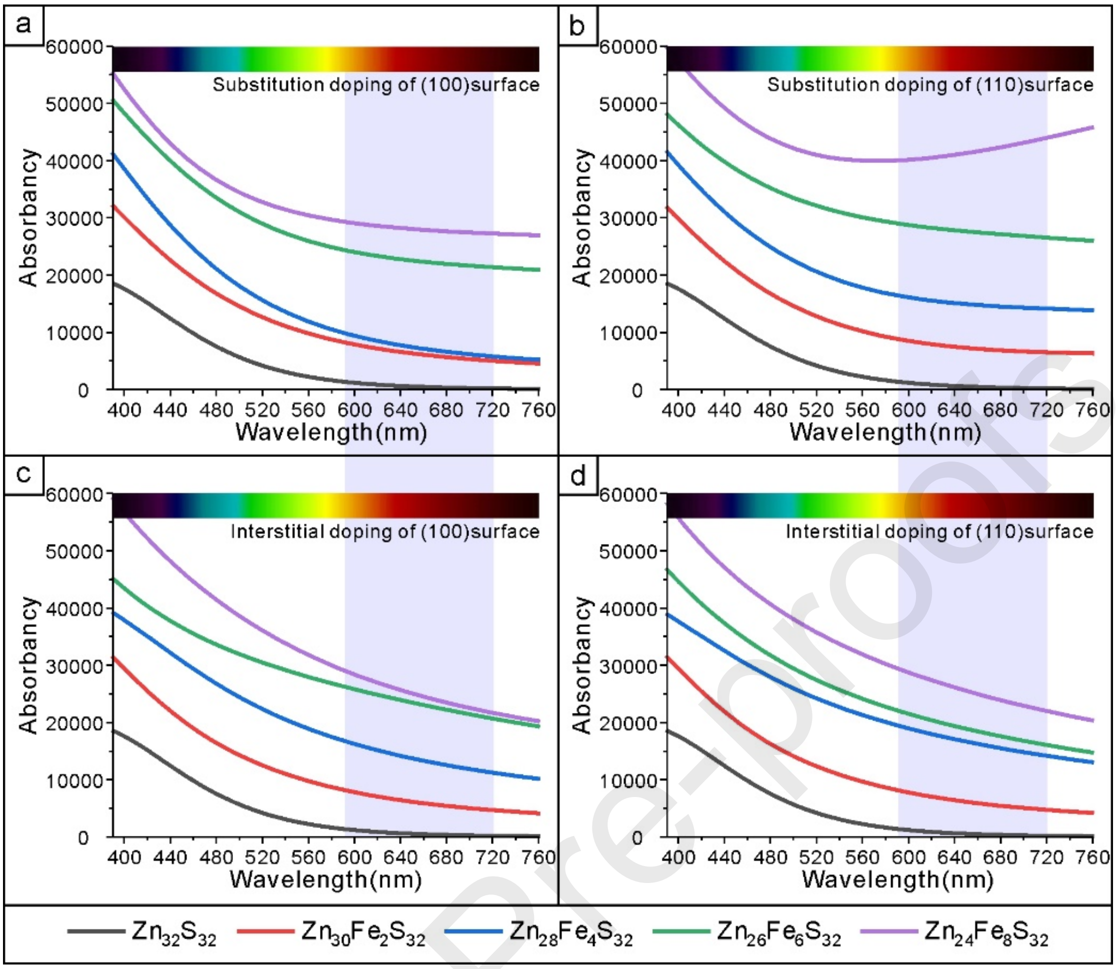

Fig. 8. Variation of absorption coefficients of Fe into the lattice interstitial solvation surfaces (100), (110) in the form of mass-like isomers and interstitial atoms.

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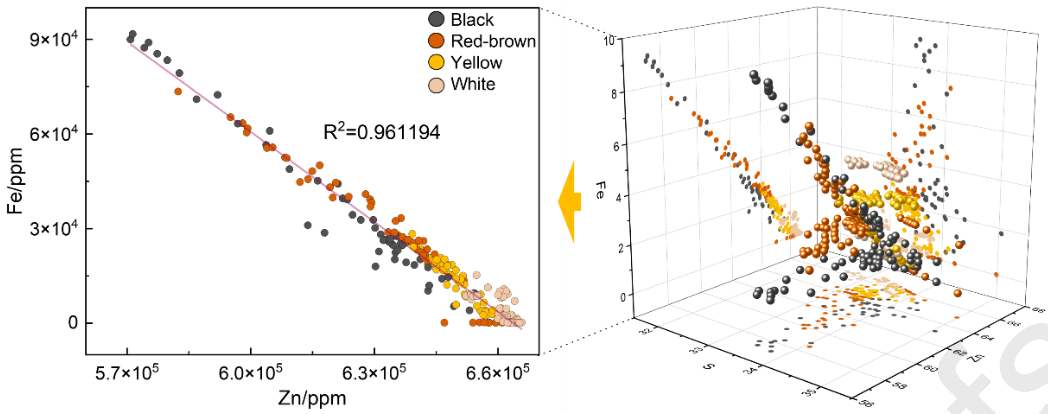

Fig. 9. Illustration of Zn–Fe correlation of different color sphalerite in Huize Pb–Zn deposit.

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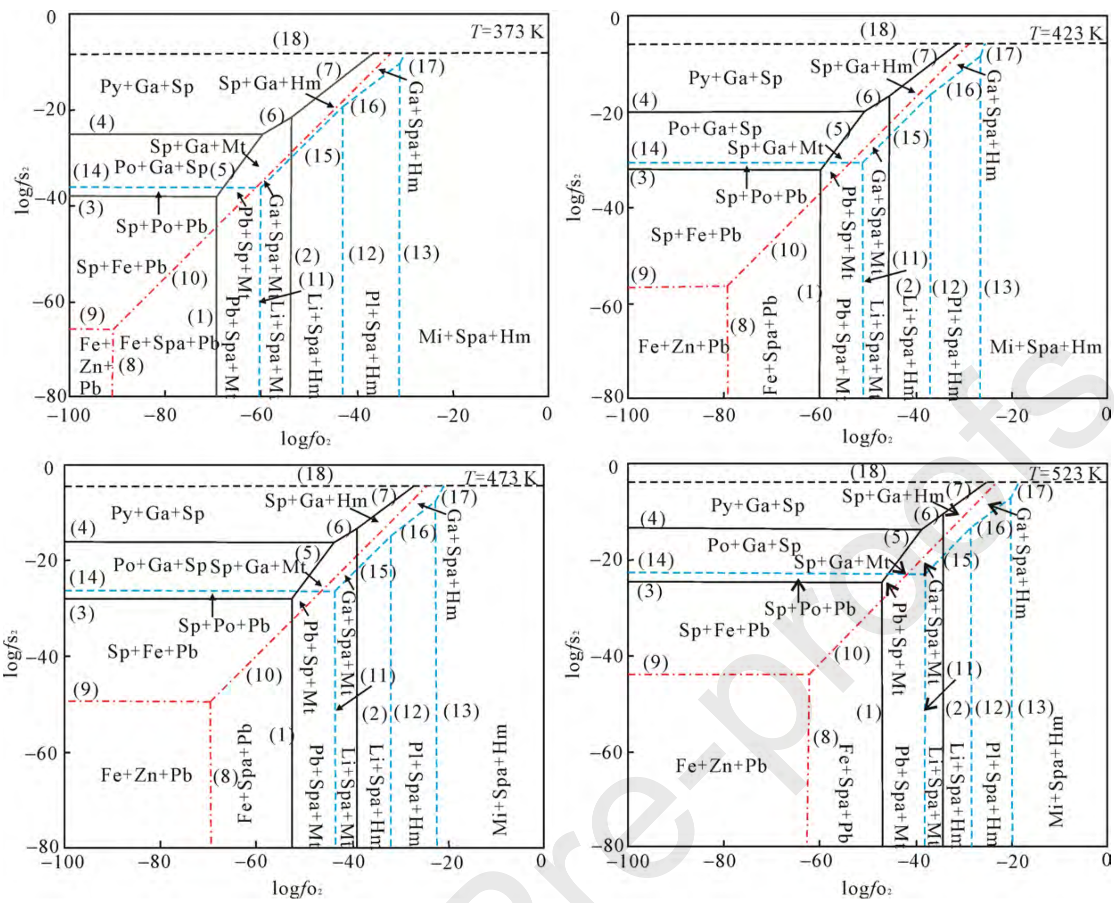

Fig.10. The $\log fO_2$–$\log fS_2$ phase diagrams at different temperatures(Zhang et al., 2014)

Declaration of Interest Statement

Article Title: Coloration mechanism of Fe in sphalerite based on DFT and LA-ICP-MS: A case study of the Huize super-large Ge-rich Ag–Pb–Zn deposit in Sichuan–Yunnan–Guizhou

Authors: Yanhong Liu, Runsheng Han, Yan Zhang, Yi Chen, Jianbiao Wu

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4. All authors read and approved the final manuscript.

5. All authors declare no conflict of interest.

Highlights

1.LA-ICP-MS and DFT calculations revealed Fe-Zn substitution effects on sphalerite color.

2. Fe substitutes preferentially in sphalerite's (110) cleavage plane over (100), forming isomorphous mixed crystals.

3.Mineralization temperature variations control sphalerite's Fe content, governing its coloration.

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